Canine Oral Microbiome

HUMAnN functional pathway analysis: detailed methods, findings, and figure atlas

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Purpose. Extend the current MetaPhlAn manuscript by characterizing age-associated changes in microbial pathway potential and their assigned taxonomic contributors.

Data. October 15 HUMAnN pathabundance/pathcoverage tables, 695 sample records with age, 626 with weight. Metadata do not include clinically measured periodontal disease severity.

Important QC issue. The table contains 695 unique samples; Madeline reported n=625 after a 2-million-read cutoff. The input must be reconciled against the read-count manifest before manuscript inclusion.

Cluster integration. Package prepared for the existing Hoffman2 Slurm-preview analysis at /u/home/z/zyjiang/canine_oral_microbiome_repro/pipeline/humann_functional. It has been locally rerun and reproducibility-checked; it has **not** been uploaded or executed on the remote cluster.

At-a-glance outcomes

Metric	Exploratory result
Sample count	695, not verified against 625 emailed
Pathways filtered at >=10% prevalence	309 / 458
Age associations, raw CPM BH q<0.05	257 / 309
Concordant/significant across four screens	136 / 309
Robust screen + pathway coverage >=0.5 in >=30%	22 candidates (8 positive, 14 negative)
Histidine degradation I vs age	Spearman rho +0.498; BH q=8.99e-44
CLR PCA PC1 vs age	Spearman rho +0.574; PC1 variance 14.9%

These results are exploratory and do not establish periodontal disease associations, pathway activity, or causation.

1. Scientific context and workflow

Our original study describes age-associated microbiome community restructuring, including higher abundance of Porphyromonas/Tannerella and lower abundance of Conchiformibius/Neisseria in older dogs. HUMAnN addresses a distinct question: which **gene-encoded metabolic pathways** shift with age, and which species are assigned to those pathways?

1.1 Files and cohorts

The community-level pathway abundance table contains MetaCyc pathway CPM estimates. The pathway coverage table provides 0-1 evidence scores for pathway representation; coverage is not an abundance normalization. Both files also contain genus/species stratified entries. The analysis excludes stratified columns from community tests and uses them only for contributor decomposition.

Observed: 695 unique dogs; 695 ages; 626 weights; 533 dogs aged at least one year. Age groups: <1 year (162), 1-<3 years (184), 3-<7 years (184), and 7+ years (165). Since raw read depths are absent, the 2-million-read eligibility cannot be independently confirmed.

1.2 Filtering and primary age analysis

Retain community MetaCyc pathways detected at nonzero CPM in at least 10% of dogs (309 of 458). For each pathway, compute Spearman rank correlation with age; adjust the 309 two-sided p values using the Benjamini-Hochberg false-discovery-rate procedure. This describes monotonic association, not an effect size per year or a causal relationship.

1.3 Compositional and age-range sensitivity

Transform filtered features with a +1 CPM pseudocount: $clr_{ij} = \ln(CPM_{ij}+1) - mean_k[\ln(CPM_{ik}+1)]$. Rerun age Spearman/BH using CLR values. Separately rerun raw and CLR tests on dogs aged >=1 (n=533). Because the data are compositional and zero-inflated, CLR is a sensitivity analysis, not a full correction of measurement bias.

1.4 Weight-adjusted models

Among dogs with weight (n=626), separately fit HC3 robust OLS: ln(CPM+1) ~ age + ln(weight+1), and the analogous CLR outcome. Test the age coefficient, then BH-adjust 309 p values. These are linear approximations without breed, sequencing depth, batch, or periodontal diagnosis.

2. QC, ordination, coverage, and taxonomic attribution

2.1 Sample-level and functional ordination

The sum of community-level CPM is close to 1,000,000 in each sample. Compute the number of detected pathways per dog and test its Spearman association with age. Perform unsupervised PCA of CLR abundance, color dog scores by age, and correlate PC1 with age. PCA is descriptive; the sign of principal components can reverse between computing environments.

2.2 Coverage-supported prioritization

Per pathway, compute the fraction of dogs with HUMAnN coverage >=0.5 and >=0.9, mean coverage, nonzero prevalence, and abundance-coverage agreement. A pathway is shortlisted when it is significant, has concordant age direction in raw CPM, CLR, adult-only CPM, and weight-adjusted CPM, **and** has coverage >=0.5 in at least 30% of dogs. These cutoffs are explicit exploration heuristics and should not be represented as HUMAnN-validated diagnostic thresholds.

2.3 Species attribution

Using stratified columns matching PATHWAY|g__...s__..., sum CPM by species across dogs. Report (a) the percentage of all species-attributed pathway CPM assigned to each species and (b) the total species-attributed CPM divided by total unstratified community CPM. Because unassigned/attribution gaps exist, the two denominators must not be conflated.

2.4 Functional category annotation

The current code labels pathways into broad keyword-defined groups for exploration only. These are **not** curated MetaCyc superclasses, and Figure 05 should not be described as official pathway-class enrichment. A bundled R script exports the file2meco curated MetaCyc mapping, requiring a mapping audit before biological claims.

2.5 Main caveats

The analysis measures encoded pathway **potential**, not RNA expression, biochemical flux, or metabolite concentration. Strong correlations do not establish disease status; the cohort lacks periodontal examination data. Broad age associations might also reflect compositional effects, correlated pathway definitions, sequencing coverage, host body size, and technical factors.

3. Statistical results and sensitivity

Analysis	Sample size	Significant pathways (BH q<0.05)
Age vs raw CPM	695	257 / 309
Age vs CLR	695	242 / 309
Age vs raw CPM in adults	533	219 / 309
Age vs CLR in adults	533	207 / 309
HC3 age coefficient adjusting for weight	626	222 / 309
HC3 CLR age coefficient adjusting for weight	626	225 / 309
Significant same-direction in 4 checks	Various	136 / 309

These counts should **not** be interpreted as 257 independent biological processes. MetaCyc pathways often share enzymes and precursor molecules, and CPM is compositionally closed.

Community-level context

PC1 of the filtered CLR pathway abundance matrix explains 14.9% of variance and correlates with age (rho +0.574). The number of pathways detected has age correlation +0.324. Because original sequencing depth is unavailable, neither the richness correlation nor broad pathway shifts should be overinterpreted.

Weight associations

Weight is recorded for 626 dogs. Significant associations with weight: raw Spearman 207/309, CLR Spearman 186/309; age-adjusted HC3 weight coefficients remain significant in 187/309 raw log1p and 181/309 CLR models. Weight and age themselves are correlated (rho +0.131, p=0.001). Breed/body size and read depth remain uncontrolled.

Technical completeness warning

181/458 community pathways have zero coverage in every dog. **135/309** tested pathways never reach coverage >=0.5, despite **114** of those being age-associated in the unadjusted raw CPM screen. Therefore coverage-backed leads should take precedence over highly significant but weakly reconstructed pathways.

4. Metabolic findings to discuss with the team

Pathway	Age rho / BH q	Coverage >=0.5	Interpretive priority
Histidine degradation I	+0.498 / 8.99e-44	50.2%	Highest-priority mechanistic lead
Histidine degradation III	+0.521 / 2.35e-48	12.9%	Lower-coverage secondary lead
Anaerobic purine degradation I	+0.515 / 2.99e-47	~0%	Association only; pathway uncertainty
Lysine fermentation	+0.400 / 1.86e-27	0%	Do not claim complete process
Denitrification	+0.326 / 3.02e-18	0%	Do not claim complete process
Fucose degradation	-0.530 / 2.73e-50	6.0%	Young-age substrate-use lead
Gondoate biosynthesis, anaerobic	-0.469 / 2.47e-38	95.0%	Counterexample to simple oxygen narrative
UMP biosynthesis III	+0.538 / 8.81e-52	73.4%	Well-covered biosynthesis lead
L-valine biosynthesis	-0.524 / 6.41e-49	73.7%	Well-covered young-age lead

Histidine degradation I: species contributions

67.9% of cumulative HISDEG-PWY community abundance is attributed to named species. Within this taxonomically attributed portion, Porphyromonas species collectively account for approximately 69.8%, and Tannerella forsythia for 4.5%. Individual assigned shares: P. gulae 29.5%, P. cangingivalis 24.7%, Bergeyella zoohelcum 17.3%, P. gingivicanis 12.2%. These shares use the **attributed-species** denominator.

How it connects to the taxonomic story

A Porphyromonas-associated rise in histidine degradation potential provides a functional complement to the age-associated increase in periodontal-linked taxa described by the MetaPhlAn analysis. It does **not** demonstrate periodontal disease or histidine-catabolite production. Conversely, decreases in mixed-acid fermentation and anaerobic gondoate biosynthesis show that one cannot say all anaerobic pathways increase with age.

5. Planned manuscript integration and decisions

Near-term result: Prioritize a new functional-analysis panel or supplement rather than immediately replacing CAZy. Compare age-associated pathways with coverage evidence, show representative age trends, and identify species contributions after exact sample-ID joining to MetaPhlAn.

Needed before publication

1. Obtain Madeline's >=2-million-read QC manifest and resolve the discrepancy between 695 actual samples and 625 reported. Recompute if exclusions are needed.

2. Reconcile pathway samples with the manuscript's earlier taxonomic QC1-QC4 cohorts; never compare unmatched sample sets without indicating that limitation.

3. Obtain sequencing depth, batch/plate, breed/sex and, if available, periodontal examination severity; perform explicit covariate-adjusted and sensitivity models.

4. Export and audit official MetaCyc categories; use eggNOG/KO gene-family data to validate mechanistic pathways once Madeline releases it.

5. Compare HUMAnN with CAZy using identical dogs, an identical primary endpoint and fair statistical controls; do not judge strength by p-values alone.

6. If age prediction is a formal goal, use nested, subject-level held-out cross-validation for functional, taxonomic and combined models.

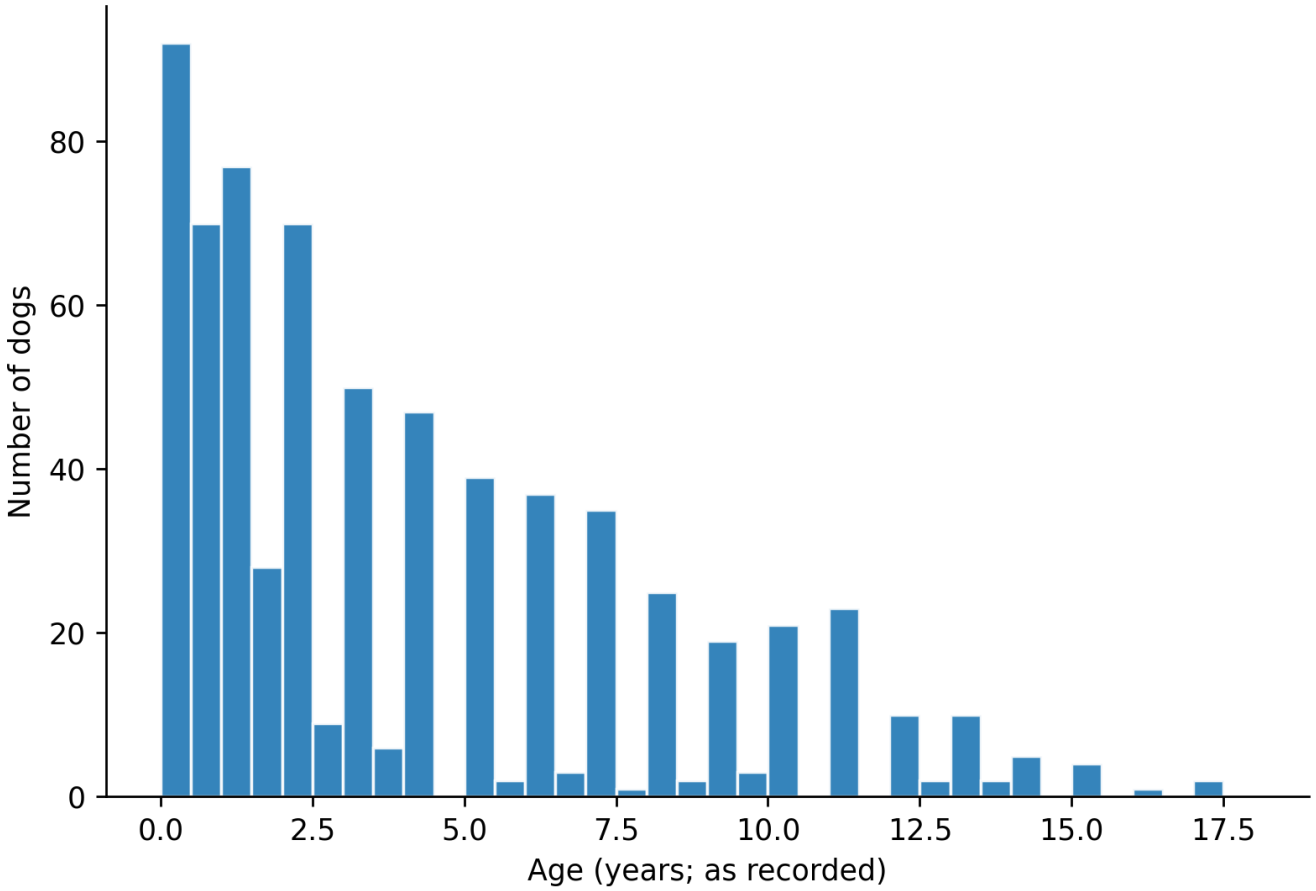
Hoffman2 implementation

The cluster-ready tarball places TSVs, analysis code, reference results, figures, and submission scripts under the isolated humann_functional directory. sbatch submit_humann.slurm writes into cluster_runs/slurm_JOBID, preserving the original microbiome pipeline. SHA256 sums verify transfer integrity, while reference_comparison.json audits rerun headline values.

Authentication: Direct remote upload or job submission has not occurred from this session. Run deployment/upload_from_mac.sh through the authenticated zyjiang SSH login. The included instructions intentionally do not store passwords or private keys.

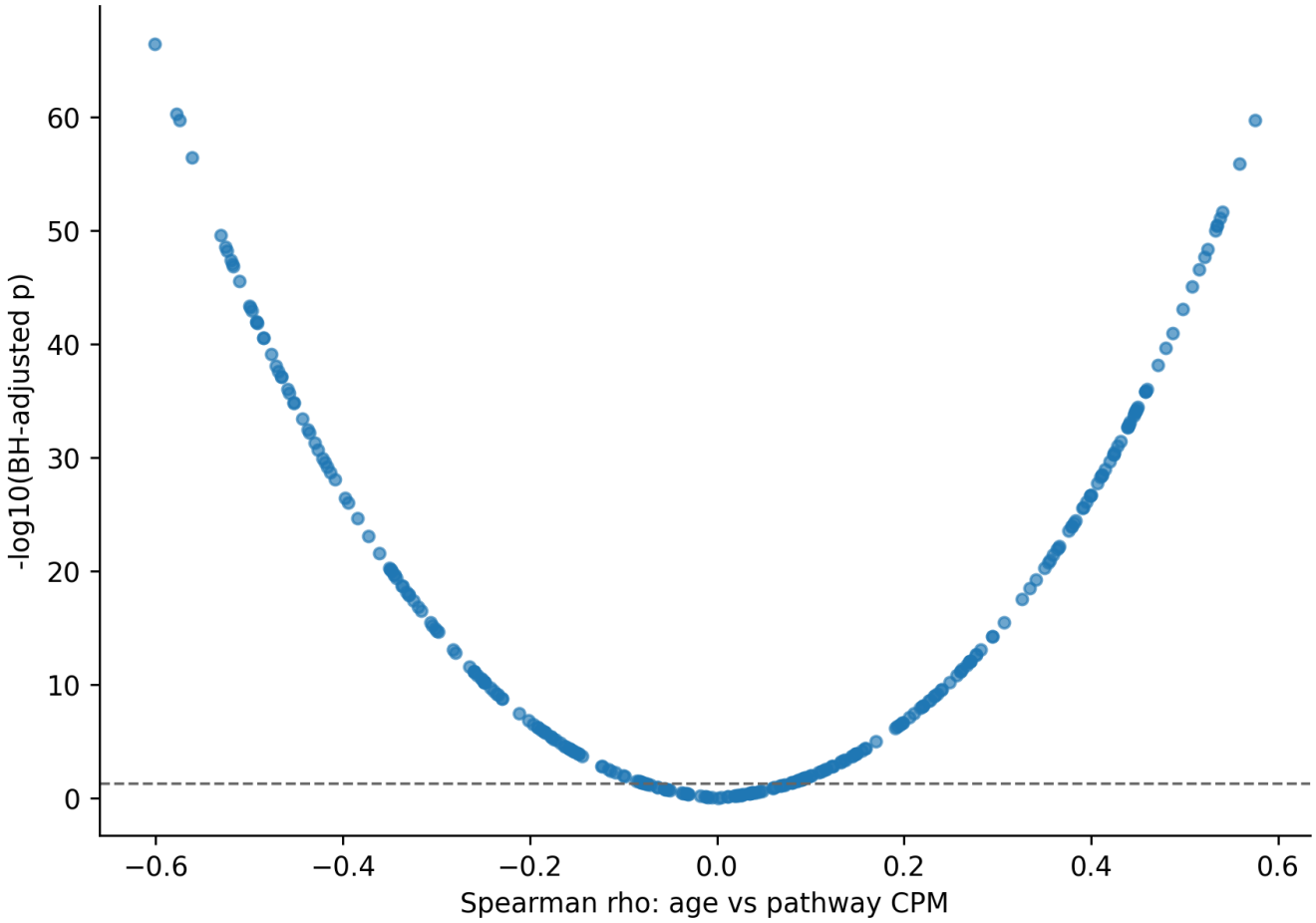
Figures 01-12 follow. Figure 05 uses keyword groupings and is not a validated MetaCyc hierarchy plot. See reports/METHODS_AND_RESULTS.md for the full protocol and reports/FIGURE_CAPTIONS.md for all expanded captions.

Figure 01: 01 age distribution



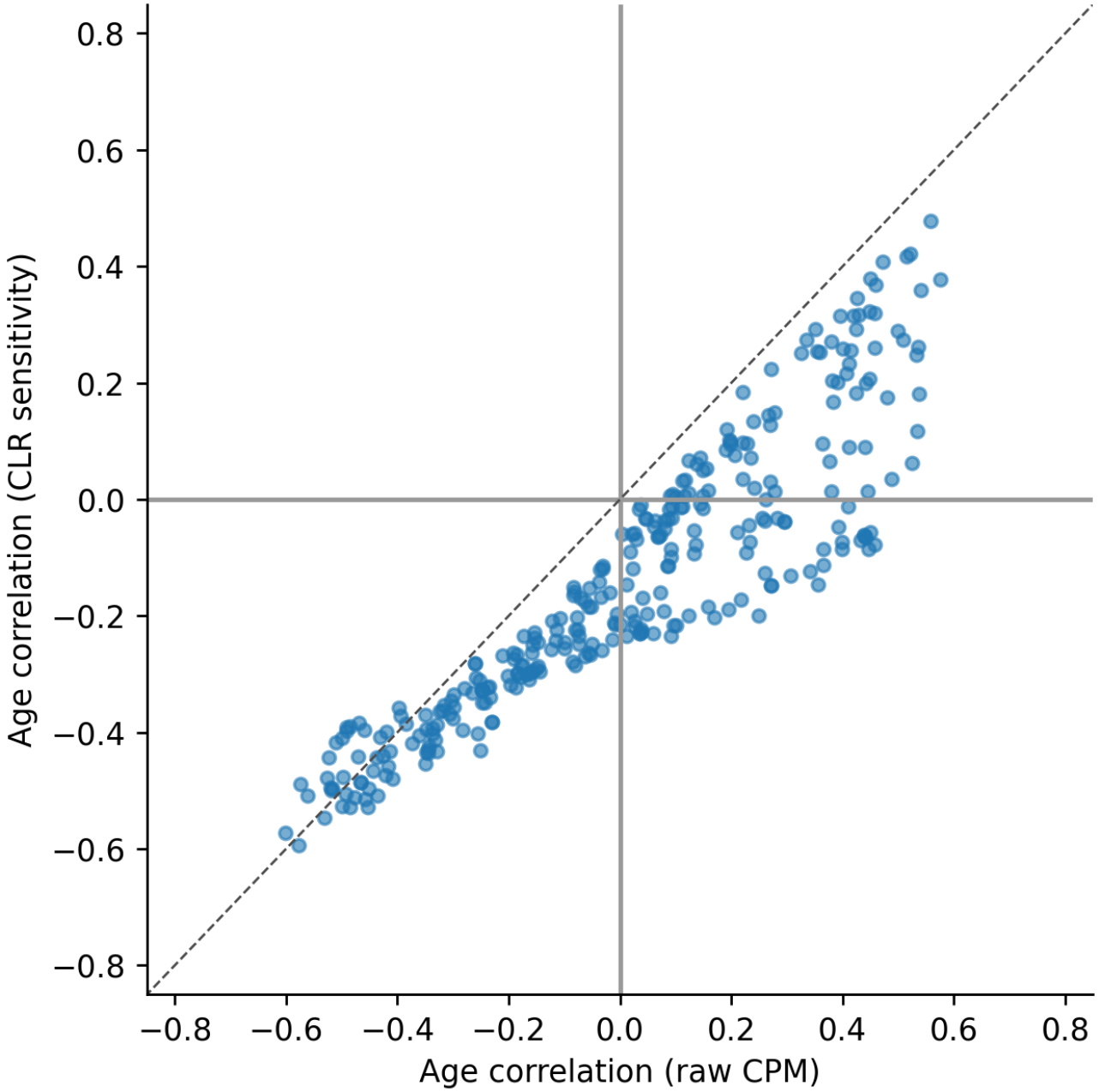
Age distribution of the 695-sample HUMAnN cohort; validate the depth-filter sample manifest.

Figure 02: 02 pathway age volcano



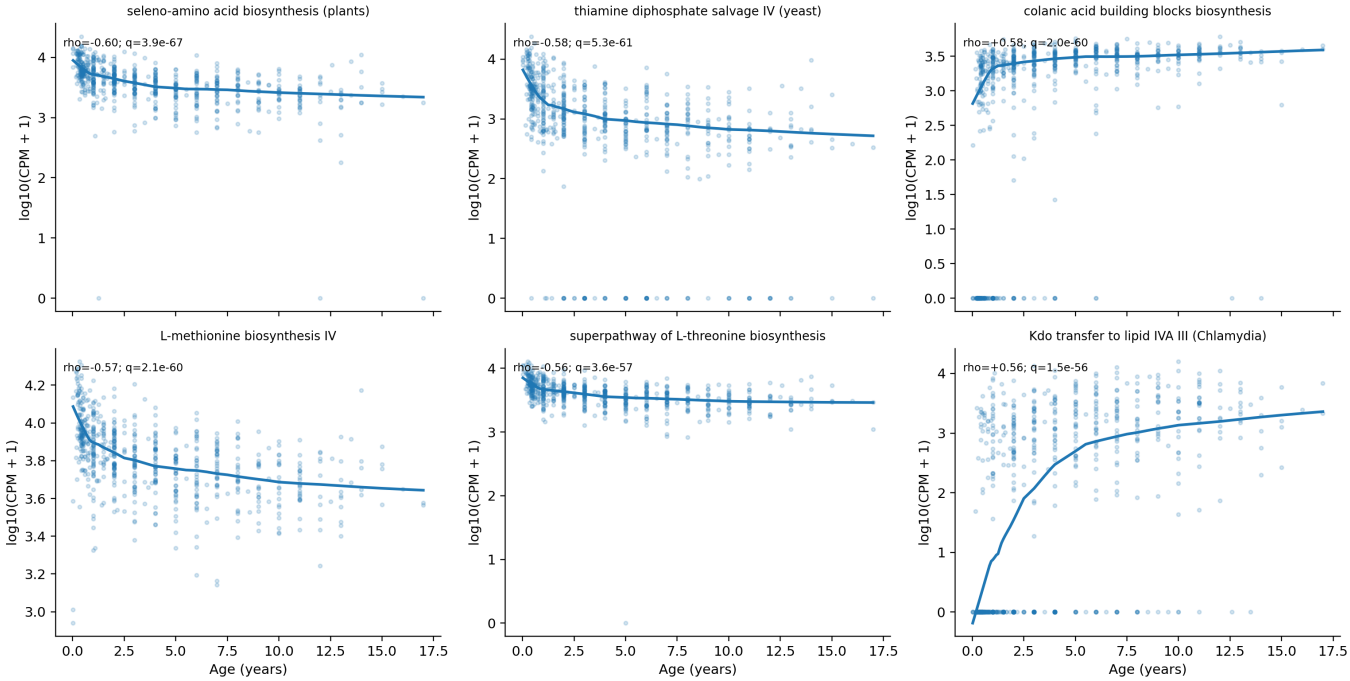
Association strength and FDR across 309 pathways. Large pathway dependence limits independent interpretation.

Figure 03: 03 raw vs clr age correlations



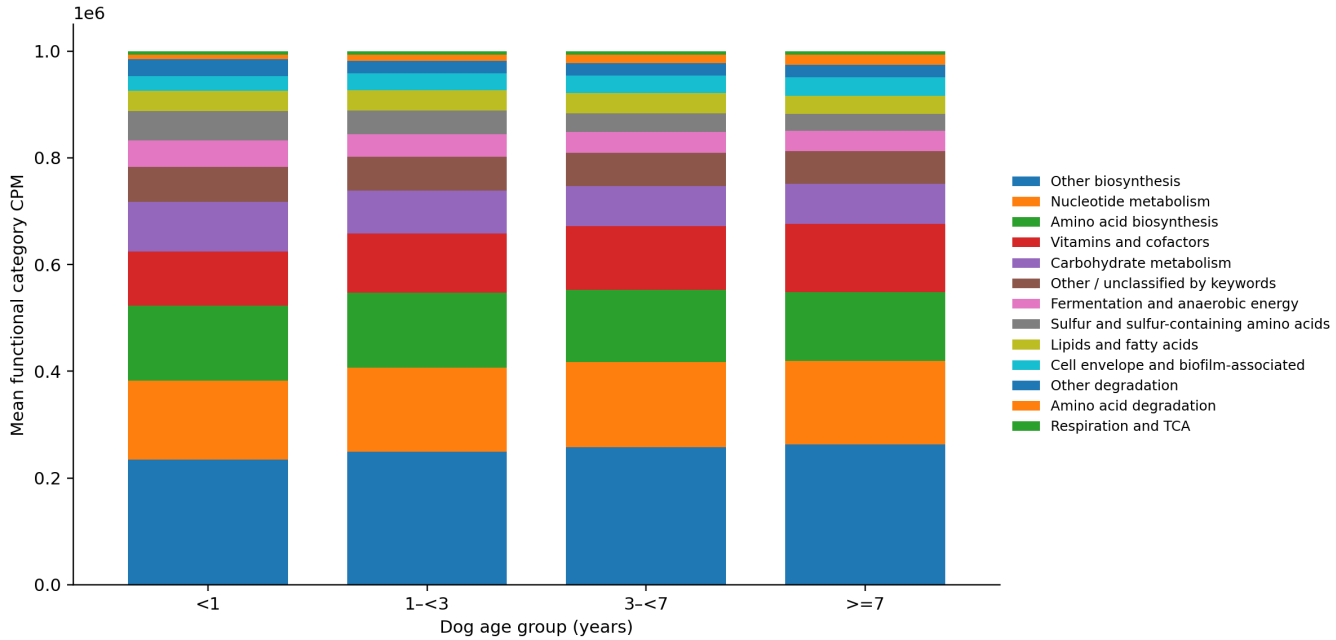
The raw CPM and CLR rank-correlation sensitivity checks show broad directional agreement.

Figure 04: 04 top robust pathways vs age



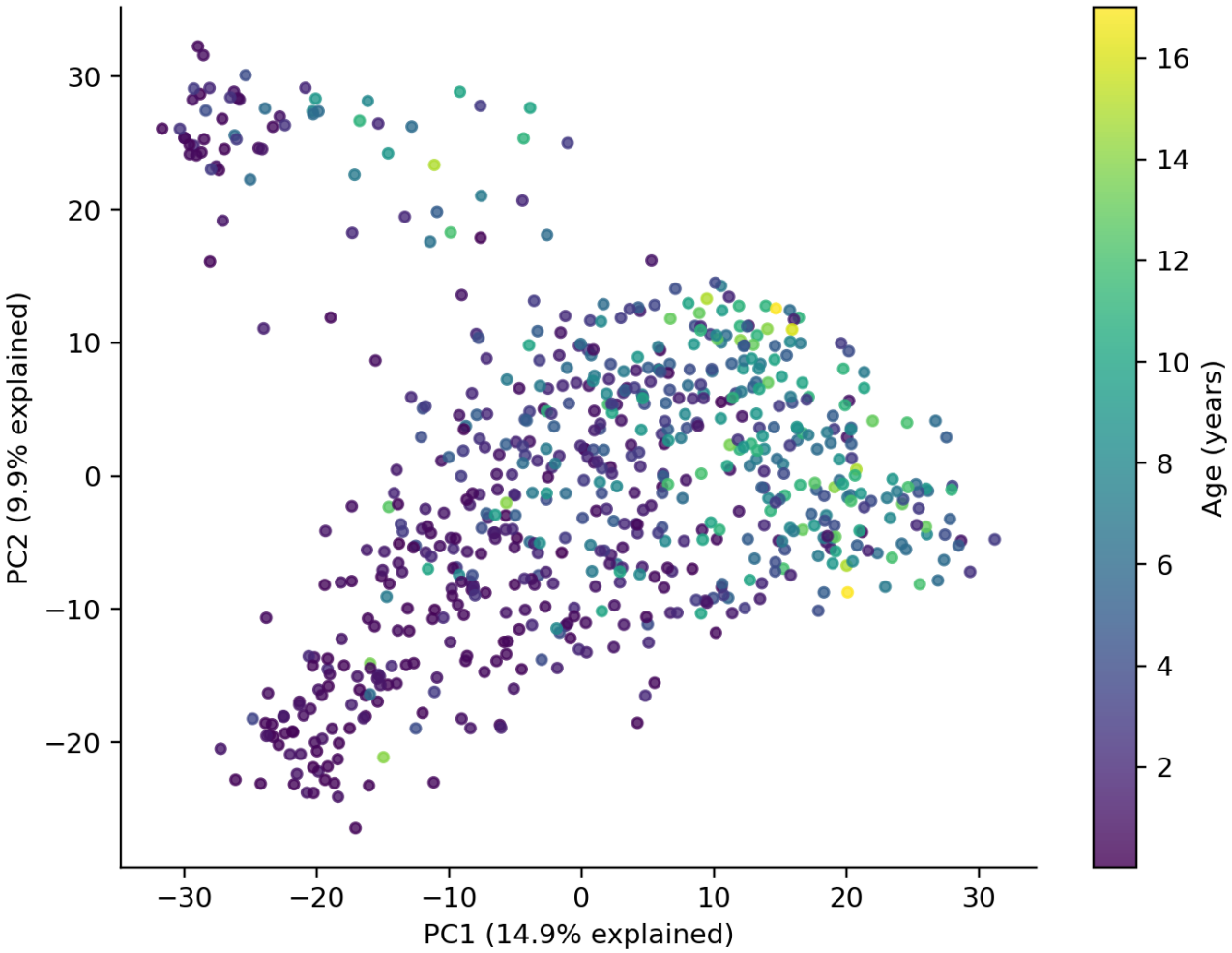
Selected robust pathway abundance-age relationships, log(CPM+1), with LOWESS smoothing.

Figure 05: 05 provisional functional categories by age



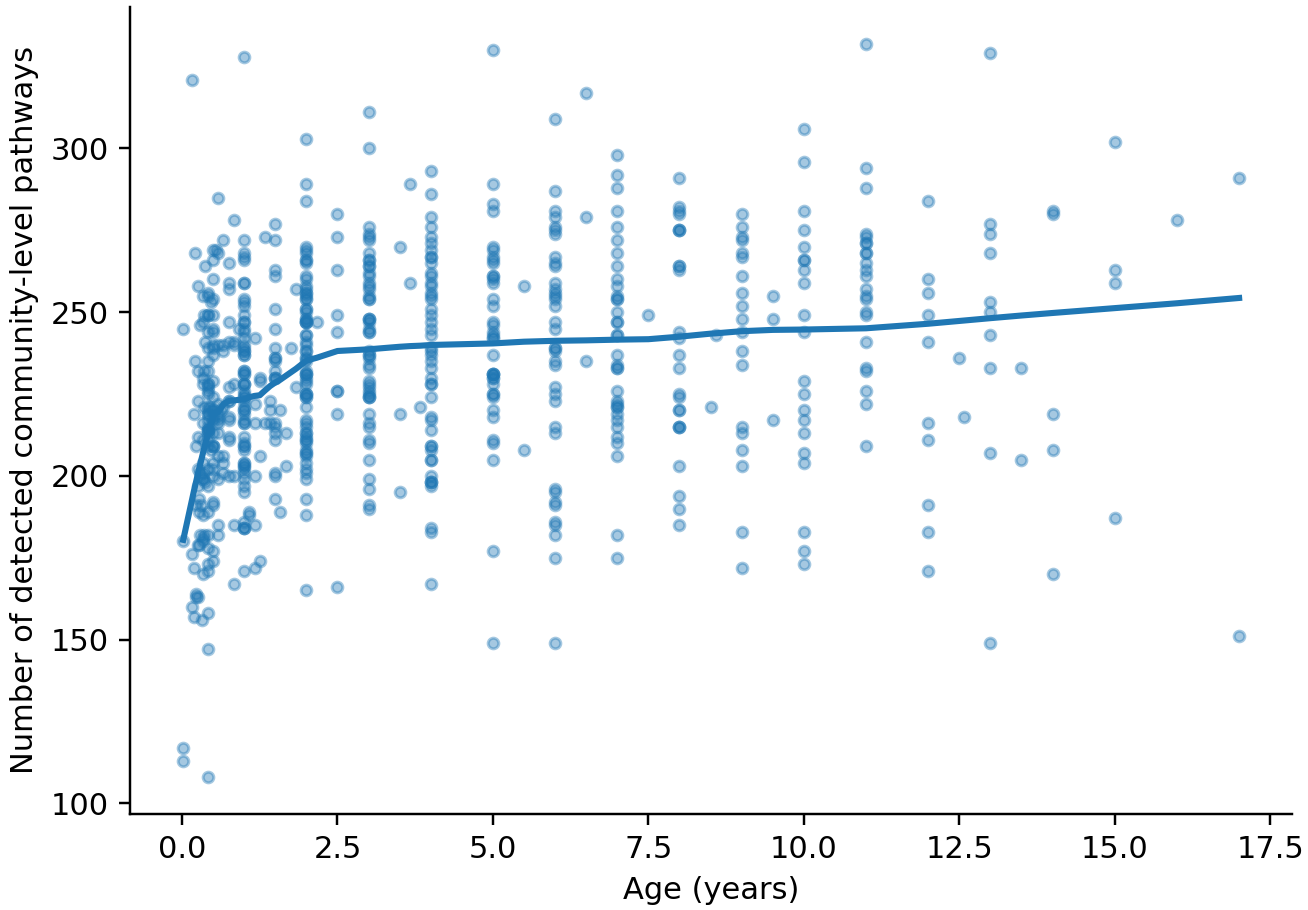
Keyword-derived categories only, not official curated MetaCyc superclasses.

Figure 06: 06 clr pathway pca by age



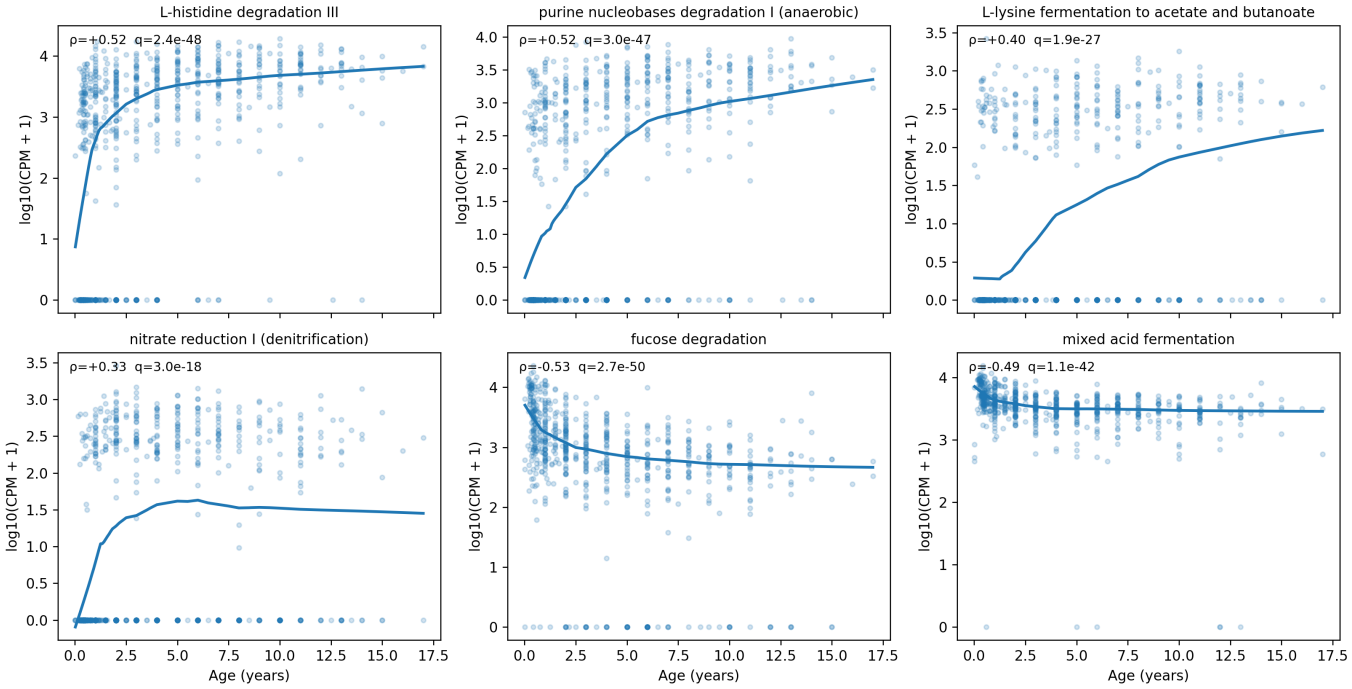
Unsupervised CLR pathway PCA colored by age; PC1 explains approximately 14.9% of variance.

Figure 07: 07 pathway detection vs age



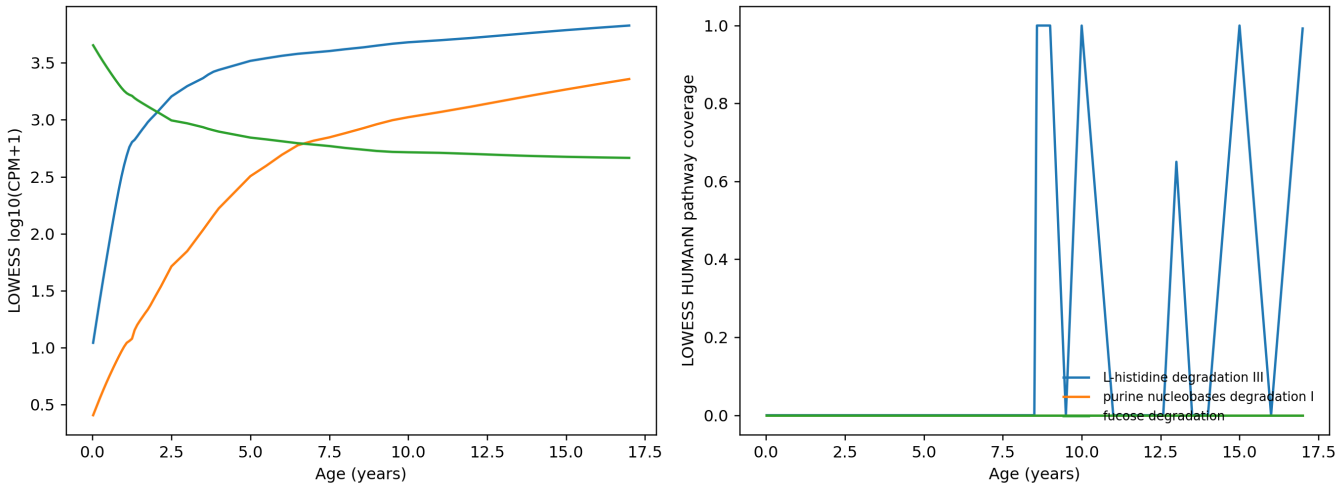
Detected pathway count rises with age; interpret cautiously without sequencing-depth controls.

Figure 08: 08 selected metabolic hypothesis pathways



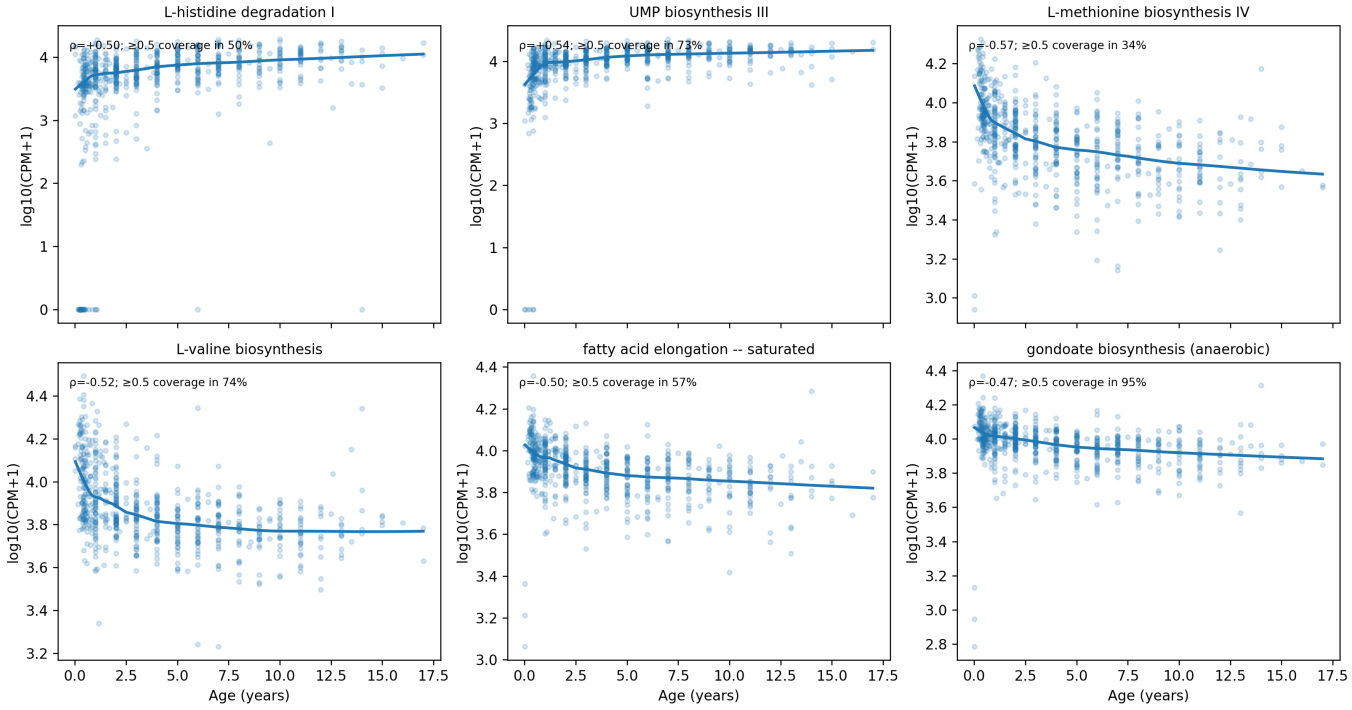
Biologically motivated candidates: several have weak pathway-coverage support.

Figure 09: 09 selected pathway abundance vs coverage



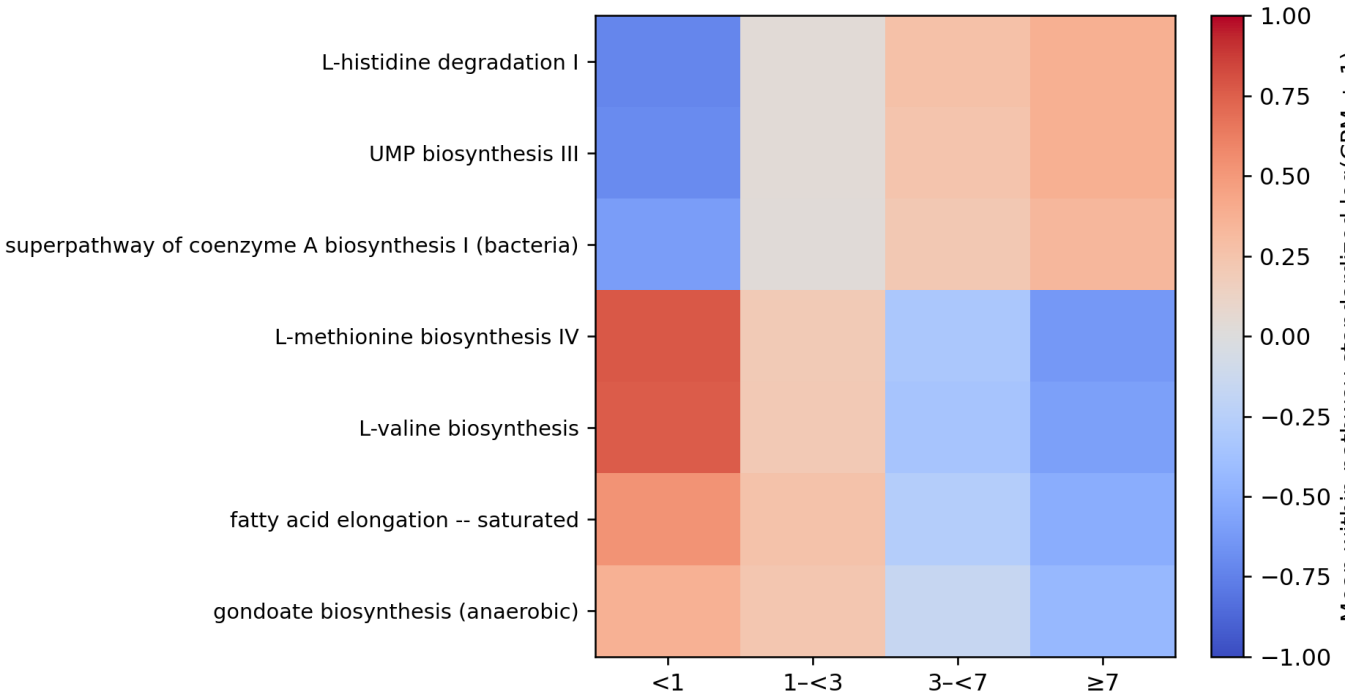
Direct comparison illustrating why HUMAnN abundance and coverage need separate scrutiny.

Figure 10: 10 coverage supported age pathways



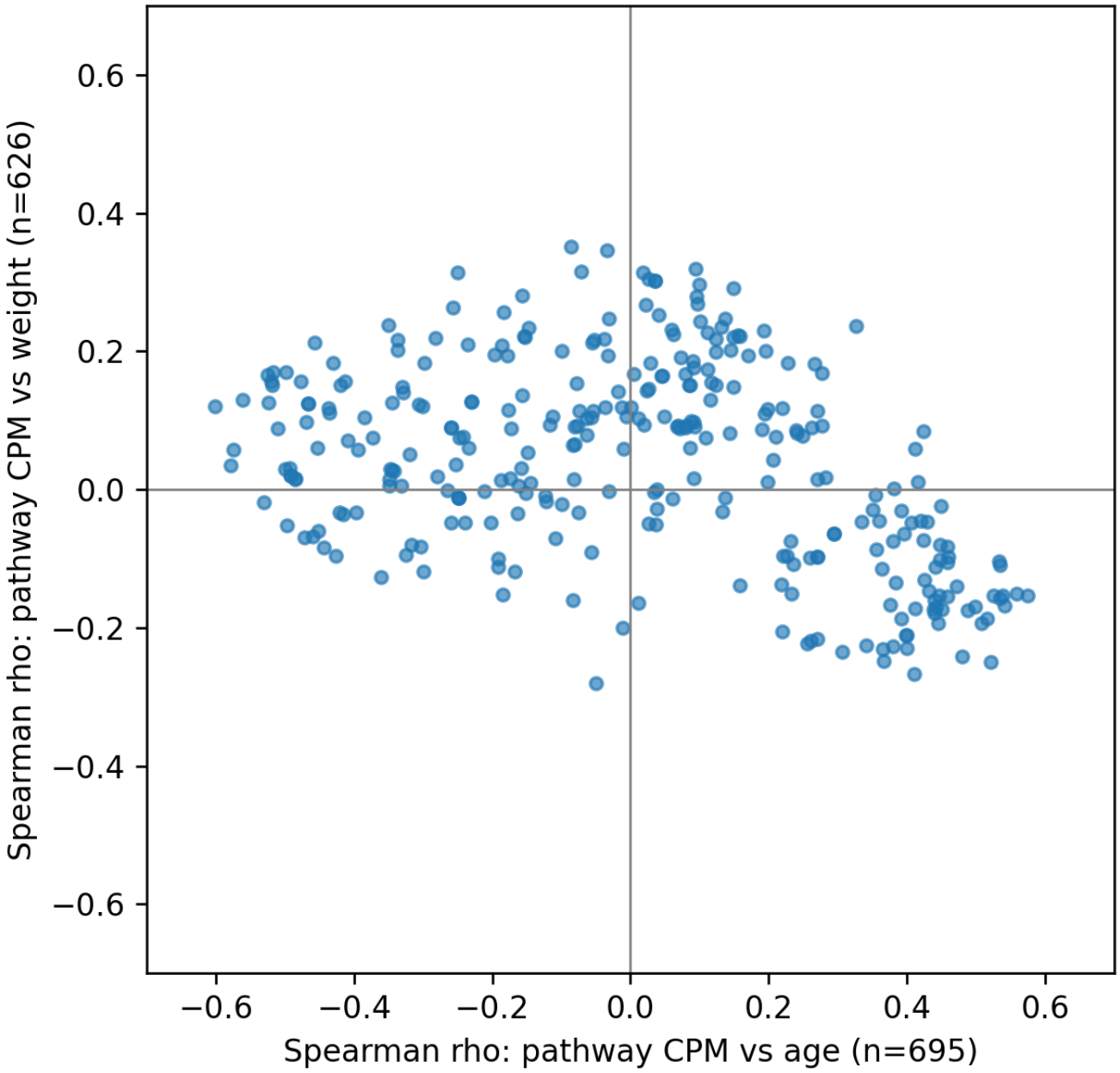
Priority pathways with stronger coverage support, including histidine degradation I.

Figure 11: 11 age group supported pathway heatmap



Age-binned standardized pathway abundances across selected well-covered pathways.

Figure 12: 12 age vs weight pathway correlations



Comparison of age and weight associations; weight analyses involve fewer dogs.